UNIT 1 · 8 hours

Control System Components



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
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Learning objectives

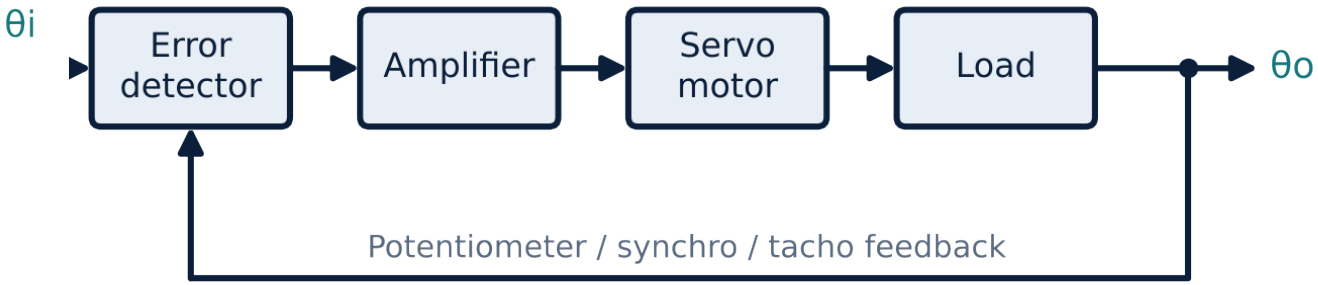
- Name the main components of a servo system
- Explain error detectors: potentiometer pair and synchros
- Describe DC and AC servo motors and the tachogenerator
- Identify the role of amplifiers and gear trains

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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A typical position servo



- Error detector compares reference and output.
- Amplifier boosts the small error signal.
- Servo motor (actuator) drives the load.
- Feedback element (potentiometer, synchro, tacho) measures the output.

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Error detectors

Potentiometer pair

Two potentiometers across the same supply; output voltage ∝ (θi − θo). Low-cost, DC systems.

Synchro pair

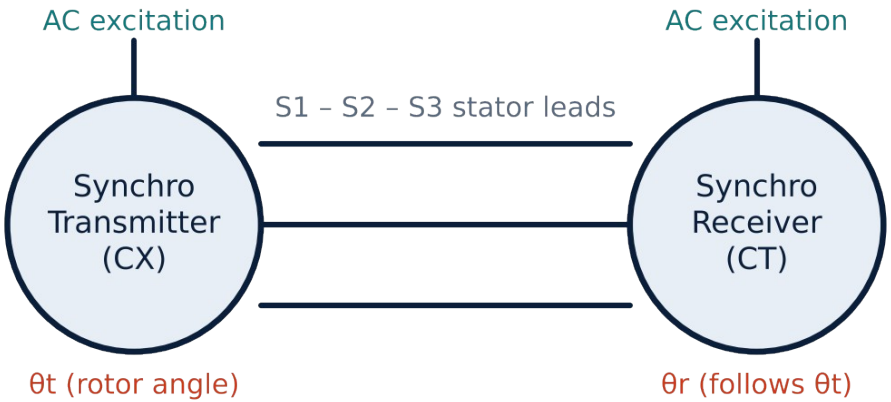
Transmitter (CX) and control transformer (CT) give an AC error voltage ∝ sin(θi − θo); accurate, long-distance.

Others

Optical encoders, LVDT, strain gauges, op-amp summers.

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Synchro transmitter and receiver



Synchro pair as an error detector

- Rotor of the transmitter turned by θt.
- Stator voltages induced in S1, S2, S3 depend on θt.
- Receiver/CT rotor at θr: output ∝ sin(θt – θr).
- Zero output when θt = θr (aligned).

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Actuators: servo motors

Feature	DC servo motor	AC servo motor
Supply	DC (armature or field controlled)	Two-phase AC: fixed phase + control phase
Torque–speed	Linear for armature control	Made linear by high rotor resistance
Typical size	High power	Low power (≤ ~100 W)
Maintenance	Brushes/commutator	Rugged, no brushes
Uses	Robots, machine tools	Instrument servos

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Tachogenerator and other components

$$e_o = K_t \omega = K_t \frac{d\theta}{dt}$$

tachogenerator output

$$E_o(s)/\Theta(s) = K_t s$$

transfer function

Gear ratio N1/N2

reflects J, B and torque

- Tachogenerator: speed feedback gives damping (rate feedback).
- Amplifier: electronic (op-amp), magnetic amplifier, amplidyne.
- Gear train: matches motor speed to load, multiplies torque.

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Key takeaways

- Servo = error detector + amplifier + actuator + load + feedback element
- Potentiometer pair and synchro pair give the error signal
- AC and DC servo motors act as actuators
- Tachogenerator provides velocity feedback: $e_o = K_t \omega$

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Practice questions

Q1. Draw and label the block diagram of a position servo system.

Q2. Explain how a synchro pair works as an error detector.

Q3. Compare AC and DC servo motors.

Q4. What is the function of a tachogenerator?